

The Gauge Algebra of Conformal Gravity

A Direct Check on the Bach Noether Identities

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Abstract

The Weyl action $S = \alpha \int \sqrt{-g} C^2$ is invariant under diffeomorphisms and under local Weyl rescalings. That much is a calculation. Whether $\text{Diff} \ltimes \text{Weyl}$ is *all* of its gauge algebra is a different question, and a harder one, since it asks about symmetries the action was not built to have.

This question has an answer, and it is not ours. Boulanger and Henneaux [6] showed by cohomological deformation theory that the Weyl action is the only consistent deformation of linearized conformal gravity with no more than four derivatives of the metric, and—the part that bears on us—that diffeomorphism invariance *follows from consistency* rather than being imposed. On the question of principle we add nothing.

What we add is a direct verification by an unrelated route. By Noether’s second theorem gauge generators and Noether identities are the same data, so the question becomes whether the Bach tensor satisfies any local identity beyond $\nabla^a B_{ab} = 0$ and $g^{ab} B_{ab} = 0$ —a question about the *full nonlinear* action, asked order by order in the gauge parameter, rather than about deformations of the free theory. Working over a metric family carrying symbolic parameters, so that a vanishing result is zero *as a polynomial* rather than at sampled fixtures, we find no further generator at derivative order zero, one or three in the gauge parameter, at the curvature degrees stated. Orders two and four are empty for a reason that needs no computation at all: index parity. Order three is the delicate case, because the third covariant derivative of the Bach tensor is seventh order in the metric and most of an affordable jet is truncation artefact; we measure how much is usable rather than assume it, and settle the case by exhibiting a rank that does not move as the usable range triples.

Two features of the problem make a naive count wrong, and both bite. Gauge symmetries form a module over *functions*, so a generator multiplied by a scalar is not a second generator; we find the trap is worse than that, since $g_{\mu\nu}(\nabla \cdot \xi)$ is the Weyl generator under a *differential* reparametrisation of its parameter, so the quotient must be by differential operators and not merely by multiplication. And every action has trivial symmetries proportional to its own equations of motion; we exclude these by construction rather than by assertion.

The result is a lower bound on completeness at the order computed, and is reported as one. We do not claim the algebra is closed at all orders. We claim that nothing hides below the order we reached, and we say exactly where that order stops.

1 The question

A gauge symmetry of an action is not a property one may simply declare. The Weyl action was constructed to be invariant under $g_{\mu\nu} \mapsto \Omega^2 g_{\mu\nu}$ and under coordinate changes, and exhibiting those

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invariances is routine. What is not routine is the converse: *is there anything else?*

The distinction matters for the same reason it matters anywhere in a reverse-physics ledger. If $\text{Diff} \ltimes \text{Weyl}$ is the whole algebra, then it is a *consequence* of the action and not an independent entry in the list of things conformal gravity assumes. If it is not, then conformal gravity has a local symmetry nobody has written down, which would be a considerably larger claim than the expected answer and would need correspondingly better evidence.

This has been answered, and the answer is not ours. Boulanger and Henneaux [6] classified the consistent deformations of *linearized* conformal gravity in $D = 4$ and found that with no more than four derivatives of the metric the only possibility is the Weyl action—and that diffeomorphism invariance is not an input but a consequence: “No a priori requirement of invariance under diffeomorphisms is imposed: this follows automatically from consistency.” On the question of principle raised above, that is the answer, and it predates this work by a quarter century.

Two things nevertheless remain to be done, and they are what this paper does. Their argument bounds the *action* by its derivative order and proceeds by deforming the free theory; ours bounds the *gauge parameter* by its derivative order and interrogates the full nonlinear action directly, through its Noether identities. The two routes share no machinery, so agreement between them is a genuine cross-check rather than a restatement. And the order-by-order form makes explicit *where* the ceiling sits, which a deformation argument does not report in those terms.

The contribution is therefore a confirmation and a bound, not a discovery. A reader who wants to know whether $\text{Diff} \ltimes \text{Weyl}$ is forced should cite [6]; a reader who wants it checked against the nonlinear action by exact computation may cite this.

Conventions. $D = 4$, signature $(-, +, +, +)$. $B_{\mu\nu}$ is the Bach tensor, and $E^{\mu\nu}$ the Euler–Lagrange expression of S , so that $\delta S = \int E^{\mu\nu} \delta g_{\mu\nu}$; for the Weyl action these agree up to a constant. All arithmetic is exact over $\mathbb{Q}$.

2 Why it is computable

Noether’s second theorem identifies gauge generators with Noether identities. A local operator $\Delta_{\mu\nu}[\varepsilon]$, linear in a parameter ε , generates a gauge symmetry exactly when $E^{\mu\nu} \Delta_{\mu\nu}[\varepsilon]$ is a total divergence for every metric — equivalently when the *adjoint* $\Delta^\dagger[E]$ vanishes identically.¹ So the question of §1 becomes:

does $B_{\mu\nu}$ satisfy any local identity beyond $\nabla^\mu B_{\mu\nu} = 0$ and $g^{\mu\nu} B_{\mu\nu} = 0$?

The two known identities are the Noether content of the two known generators: the divergence is the diffeomorphism identity, the trace is the Weyl identity. The question is whether the list is complete.

The problem splits by the *type* of the gauge parameter. A scalar parameter σ gives candidates $\Delta_{\mu\nu} = T_{\mu\nu}\sigma$; a vector parameter ξ_λ gives candidates carrying derivatives,

$$\Delta_{\mu\nu}[\xi] = S_{\mu\nu}{}^{\rho\sigma\tau\lambda} \nabla_\rho \nabla_\sigma \nabla_\tau \xi_\lambda$$

and its lower-order analogues. These are computed separately and the results combined in §4.

¹The adjoint *reverses* the order of the derivatives: the outermost derivative acting on the parameter in Δ becomes the innermost acting on E . Since the derivatives do not commute, assigning a candidate to a generator before computing the adjoint is a reliable way to turn an ordering slip into a result. We compute the contractions first and interpret afterwards.

What “over a family” means here. Every identity below is evaluated on a metric family carrying symbolic parameters, so a result of zero is zero *as a polynomial in those parameters* — true for every member at once, not at sampled fixtures. The family is general in the metric (ten parameters span every component of a symmetric 4×4) but each component carries a fixed polynomial pattern in the coordinates. It is a generic family near flat space, not every metric, and §7 says so again.

3 Two ways to overcount, both of which bite

3.1 The module is over functions, and the trap is worse than that

Gauge symmetries form a module over functions of the fields, not a vector space. If $\Delta_{\mu\nu} = g_{\mu\nu}\sigma$ is the Weyl generator, then so is $\Delta_{\mu\nu} = g_{\mu\nu}R\sigma$, and it is not a second symmetry — it is the first with its parameter reparametrised, $\sigma \mapsto R\sigma$. A kernel dimension that counts these separately overcounts, and any reported count that does not say how it handled this should not be believed.

The vector sector shows the trap is strictly worse. The degree-zero candidate $S_{\mu\nu}{}^{\rho\sigma} = g_{\mu\nu}g^{\rho\sigma}$ gives

$$\Delta_{\mu\nu} = g_{\mu\nu}(\nabla \cdot \xi),$$

which is the Weyl generator with $\sigma = \nabla \cdot \xi$: a *differential* reparametrisation of the parameter, not a functional one. So the quotient must be taken by differential operators acting on the parameter, and a count that quotients only by multiplication by functions *still* overcounts. Concretely, the naive degree-zero kernel in the vector sector has three elements and the generator count is two.

3.2 Trivial symmetries, excluded by construction

Every action, without exception, admits symmetries of the form $\Delta_{\mu\nu} = M_{\mu\nu\alpha\beta}E^{\alpha\beta}$ with M antisymmetric under exchange of its index pairs. They are infinite in number and carry no information; a count that includes them is meaningless.

They are excluded here by construction rather than by assertion. Every coefficient enumerated is built from the metric and the curvature and never from E , and the Bach tensor is fourth order in the metric, so no candidate at the degrees considered can contain it.

4 The result

Theorem 4.1 (Scalar sector). *Among candidates $\Delta_{\mu\nu} = T_{\mu\nu}\sigma$ with T built from the metric and curvature at curvature degree at most two, there is exactly **one** generator, the Weyl generator. The kernel consists precisely of the multiples of $g_{\mu\nu}$, which are reparametrisations of it; every candidate that is not such a multiple has nonvanishing adjoint.*

The candidate list for Theorem 4.1 has ten elements, not the seven of the list we inherited. Degree two has seven members in full — three scalars times g , one scalar times Ric, and three carrying both free indices on curvature — and with degrees zero and one that is ten. The three that a g -and-Ric list omits are exactly the ones contracting against the full Riemann tensor, which is the shape that could carry a second generator, since $E^{\mu\nu}$ here is the Bach tensor and sees the whole Weyl tensor rather than only its traces.

Theorem 4.2 (Vector sector). *Among candidates with a vector parameter at derivative order $k \leq 3$, the only generator is the diffeomorphism generator. Specifically:*

- $k = 0$ and $k = 1$: no further generator, at curvature degree up to two in the coefficient.
- $k = 2$ and $k = 4$: **empty by index parity**, with no computation required. A coefficient at order k carries $2 + k + 1$ indices, and every tensor built from g , g^{-1} , δ and undifferentiated curvature has even rank, so the odd-rank coefficient the even orders need cannot be built.
- $k = 3$, curvature degree zero: no further generator, established as described in Remark 4.5—a rank stable under increasing resolution, with the rank deficit accounted for by a coincidence rather than a relation. At curvature degree one the analogous sweep is **open**: its kernel numbers are not yet interpretable, because the quotient by trivial relations is currently only by pairwise equality and a scalar multiple is as much an artefact as a duplicate.

Remark 4.3 (The parity argument has a limit, which is why $k = 0$ was computed). Derivatives of curvature have *odd* rank — $\nabla_a R$ is rank one, $\nabla_a R_{bc}$ rank three — so order zero is reachable after all, with one derivative on the curvature in the coefficient. It was therefore enumerated rather than dismissed by the parity argument that disposes of orders two and four. Order zero is also the cheap sector: its adjoint is a contraction rather than a divergence, with no integration by parts.

Remark 4.4 (Scope of the degree-one sweep). Every candidate in that sweep has the form $Y = F_1[\text{pair}] \times F_2[\text{pair}]$ with *both factors of rank two*, and the enumeration is over which two of the four slots each factor occupies. It is therefore complete in *placement* within that shape class, and *not* complete in *shape*: coefficients that do not factor into two rank-two pieces—for instance Riemann carrying three free indices, $R_{\rho\sigma\tau a} B^a{}_\lambda$ —are outside it. At curvature degree one the result is consequently weaker than at degree zero, and is stated as such.

Remark 4.5 (How order three is settled, and what nearly went wrong). $\nabla^3 B$ costs seven derivatives, so most of an affordable jet is truncation artefact. Rather than trust a budget we *measure* the usable range: the contractions whose innermost operation is $\nabla \cdot B$ must vanish identically, and the highest order at which they still do says how far the data is good. The arithmetic predicted no usable orders; the measurement found three at jet degree seven, and the range grows with the jet degree—orders $0 \dots 2, 3, 4, 6$ at degrees $7, 8, 9, 10$.

Two facts then settle the case. The rank of the surviving contractions is 2 of 3 *at every one of those degrees*: a number that does not move while the usable data triples is not a resolution artefact. And the deficit is a *coincidence*—exactly one pair among the three yields the same adjoint up to sign, three vectors with one coinciding pair have rank at most two, and the rank is two. The coincidence accounts for the whole deficit, leaving nothing to be a symmetry.

An earlier version reached this conclusion by a route that does not survive scrutiny: it exhibited a candidate relation and found it failed when compared over the *whole* jet, but the failure lay entirely above the valid range. Right answer, invalid reason. We record it because the failure mode is general—“testing on the full jets” means nothing where the jet carries no information—and because the repair, measuring the usable range against a known answer before comparing anything, is the transferable part.

Corollary 4.6. *At derivative orders zero and one (curvature degree ≤ 2) and at order three (curvature degree zero), the gauge algebra of $S = \alpha \int \sqrt{-g} C^2$ is $\text{Diff} \ltimes \text{Weyl}$; orders two and four are empty by parity. This agrees with what [6] establishes by deformation theory, reached independently, over the range where our computation resolves.*

5 The negatives are the result

The positive cases in Theorems 4.1 and 4.2 are already known and are recomputed only as premises: the diffeomorphism adjoint is $\nabla^\mu E_{\mu\nu}$ and the Weyl adjoint is the trace. What makes “exactly these”

evidence rather than an enumeration that never ran is that curvature-weighted candidates *must fail*, and do.

At order one, three such candidates fail: R weighting the diffeomorphism candidate, Ric contracted into the trace slot — which requires $B^{\mu\nu} R_{\mu\nu} \equiv 0$, and that is not an identity — and Ric as a mixed factor. At order three and curvature degree one, all eight sub-families fail.

Two structural controls carry more weight than the negatives, because they are known answers rather than expectations. First, whenever the innermost derivative of a third-order candidate contracts with an index of B , the innermost operation is $\nabla \cdot B$, which is zero; those contractions must vanish identically, and do, through a rank-five object. That is the diffeomorphism identity differentiated twice, and if it failed nothing else would mean anything. Second, the contraction of Riemann’s second index pair with a symmetric tensor is zero *before any derivative is taken*, because Riemann is antisymmetric there and B is symmetric; that tests the construction rather than the adjoint machinery.

Remark 5.1 (One control is vacuous and is reported as such). The degree-zero Weyl candidate has adjoint $\nabla^\sigma(\text{tr } E)$. The Bach tensor is traceless, so this vanishes *for that reason* and not independently. It reduces to the known trace identity; it is not a check, and a control evaluated where the thing it measures degenerates proves nothing.

A complementary negative, arrived at independently

While this work was in preparation, Hobson, Barker and Lasenby [5] extended the Bessel–Hagen construction to the conformal group and proved that linearised Weyl-squared gravity admits *no* strict local, polynomial, symmetric, dimension-four rank-two tensor quadratic in $h_{\mu\nu}$ that is both gauge invariant and conserved on the Bach shell—so no strictly gauge-invariant local energy–momentum tensor exists there.

That is a different object in a different regime: a conserved *current* in the linearised theory, where ours is a gauge *generator* in the full nonlinear one. Neither result implies the other. But they are the same kind of statement about the same theory—an exhaustive search that comes back empty—reached by unrelated methods, and we record the pairing because it is the more informative reading of both. Weyl-squared gravity appears to be a theory whose structure is pinned as much by what it does not admit as by what it does.

6 Rank is a floor, and the data has an edge

The adjoint of a candidate is a linear combination of finitely many contractions, so the symmetries are that combination’s kernel. Two readings of such a computation are wrong and both were live here.

Showing each contraction separately nonvanishing does *not* show they are independent: a combination of nonvanishing objects can vanish, and that combination would be a symmetry. Conversely, a rank read at a base point is only a *lower* bound on the generic rank, so a rank deficit there is not a relation. What settles it is exhibiting the candidate combination — by carrying an identity through the elimination, so a row whose left half reduces to zero returns its own coefficients — and then testing that combination on the full jets, where a genuine relation must be the zero polynomial at every order.

This procedure has an edge, and at order three we ran off it. “Testing on the full jets” is only meaningful where the jet carries information, and after seven derivatives most of a degree-seven jet does not. A witness that *fails* above the valid range has not failed at all. That is how the withdrawn

conclusion of Remark 4.5 arose, and the guard against it is to measure the usable range against a known answer before comparing anything—which is now done, and reported alongside the rank.

Remark 6.1 (A width, not just an order). A relation may verify identically over a narrow family and still be false of the theory. One did: over a one-parameter family a relation among the R -weighted order-three candidates verified at every order, and widening the family to two parameters removed it. It was an artefact of a family too thin to separate the candidates. The parameter count is a knob to vary, not to fix at the cheapest value that runs, and the computations reported here are at the wider setting.

7 What this paper does not establish

- **Not all orders, and not order three at curvature degree one.** The result covers derivative orders zero and one, order three at curvature degree zero, and two and four by parity. Order three at curvature degree one is open—its kernel numbers are not yet interpretable, since the quotient by trivial relations is only by pairwise equality. Order five and above is untouched.
- **Not all metrics.** The family is general in the metric but each component carries a fixed polynomial pattern in the coordinates. This is a generic family near flat space. An identity that fails only on metrics far from this family would not be seen.
- **Not a proof that the ranks are full.** Only individual witnesses were tested against the full jets. A *combination* of witnesses could vanish while none does alone, so “no witness verified” is strong evidence for full rank rather than a proof of it.
- **Not complete in shape at curvature degree one.** See Remark 4.4: that sweep is exhaustive in placement within a restricted shape class, not over all coefficients of that degree.
- **No principled stopping rule for the family width.** The family carries two parameters because one produced a spurious relation (§6) and two did not. That is a repair, not a criterion: nothing here shows two is *enough*, and we do not have an argument that the family spans the jet directions the identities probe. It does not.
- **One implementation.** Unlike some results in this programme, nothing here carries an independent second rail. A systematic error in the shared curvature layer would not be caught by these checks, though the Riemann comparison in §8 narrows that exposure.
- **$D = 4$ and one signature.** Nothing is claimed in other dimensions, where the counting arguments—index parity in particular—would have to be redone.
- **No novelty in the symmetries themselves.** That the Weyl action is diffeomorphism and Weyl invariant is classical. The contribution is the converse direction — that at bounded order there is nothing else — which is what converts an assumption into a consequence.
- **Nothing about the quantum theory.** Everything here is a classical local computation. Whether these symmetries survive quantisation is a separate question with its own obstructions, treated elsewhere in this programme.

Prior art, and how hard we looked. The novelty claim of an earlier draft was wrong and was withdrawn once [6] was found; that is recorded here rather than quietly repaired. What we searched for since: prior classification of the gauge algebra of conformal gravity, prior enumeration of Noether

identities of the Bach tensor beyond transversality and tracelessness, and current work on gauge structure and conserved currents in Weyl-squared gravity. That last turned up [5], four days before this draft. We did not find a prior order-by-order search for further identities of the *nonlinear* action, which is a statement about our search and not a claim about the literature. Anyone who knows of one should regard this paper’s contribution as reduced accordingly.

8 Verification

Each result is a fail-closed certificate with a deterministic verifier, computed in Forge over exact rationals with nested jets: the outer degree carries the coordinate dependence and the inner degree the family parameters. Both truncation degrees are guarded, since a saturated jet compares truncation caps rather than objects and an exact-identity test between saturated jets is inconclusive rather than false.

The euler gate stands at 9 of 10 by design: its tenth check extends the same machinery to the conformally invariant subspace, does not pass its own known-answer control, and is therefore withheld. A gate that reported 10 of 10 while that control failed would be the more worrying number.

result	certificate	rail
Thm. 4.1	REVERSE_PHYSICS_NOETHER_IDENTITIES_V1	euler gate, 9/10
Thm. 4.2	REVERSE_PHYSICS_GAUGE_VECTOR_SECTOR_V1	family gate, 18/18
curvature inputs	REVERSE_PHYSICS_SYMBOLIC_FAMILY_V1	family gate, 18/18

The Riemann tensor these contractions are built from is pinned separately. This tree carries two implementations of it, and nothing had compared them; a contraction built against the wrong one would be wrong in a way that every symmetry test on the other one would miss. They agree componentwise, and both satisfy all four algebraic symmetries — including pair exchange, which is a theorem rather than a manifest property of the definition and which no test in this tree had previously covered.

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